Fracción Parcial

a) $\int \frac{4x+7}{x^2-3x-10}dx$

se factoriza $\rightarrow x^2-3x-10 = (x-5)(x+2)$

$$\frac{A}{(x-5)} + \frac{B}{(x+2)} = \frac{\cancel{(x-5)}(x+2)}{\cancel{(x-5)}} + \frac{(x-5)\cancel{(x+2)}}{\cancel{(x+2)}}$$

$A(x+2) + B(x-5)$

$x=5$

buscar A $\rightarrow 4(5)+7 = A(5+2) = B\cancel{(5-5)}^{0}$

$27 = A(5+2) + 0$

$27 = A(7) \rightarrow \boxed{\frac{27}{7} = A}$

$x=-2$

buscar B $\rightarrow 4(-2)+7 = A\cancel{(-2+2)}^{0} + B(-2-5)$

$4(-2)+7 = 0 + B(-2-5)$

$-1 = B(-7) \rightarrow \frac{-1}{-7} \rightarrow \boxed{\frac{1}{7} = B}$

$$\int \frac{4x+7}{(x-5)(x+2)} = \int \frac{27/7}{(x-5)}dx + \int \frac{1/7}{(x+2)}dx$$

$$\frac{27}{7}\int \frac{1}{x-5} + \frac{1}{7}\int \frac{1}{(x+2)} dx$$

$$\frac{27}{7}\ln(x-5) + \frac{1}{7}\ln(x+2) + C \,//$$

b) $\int \frac{5x+1}{x^2+3x-10}dx$ se factoriza $\rightarrow x^2+3x-10 = (x+5)(x-2)$

$$\frac{A}{(x+5)} + \frac{B}{(x-2)} = \frac{\cancel{(x+5)}(x-2)}{\cancel{(x+5)}} + \frac{(x+5)\cancel{(x-2)}}{\cancel{(x-2)}}$$

$5x+1 = A(x-2) + B(x+5)$

$x=-5$ se hace 0

buscar A $= 5(-5)+1 = A(-5-2) + B\cancel{(-5+5)}$

$-24 = A(-7) + 0$

$-24 = A(-7) \rightarrow \frac{-24}{-7} \rightarrow \boxed{\frac{24}{7} = A}$

$x=2$

buscar B $= 5(2)+1 = A\cancel{(2-2)}^{0} + B(2+5)$

$5(2)+1 = 0 + B(7)$

$11 = B(7) \rightarrow \frac{11}{7}$ $\boxed{\frac{11}{7} = B}$

$$\int \frac{5x+1}{(x+5)(x-2)} = \int \frac{24/7}{(x+5)}dx + \int \frac{11/7}{(x-2)}dx$$

$$\frac{24}{7}\int \frac{1}{(x+5)} + \frac{11}{7}\int \frac{1}{(x-2)}dx$$

$$\frac{24}{7}\ln|x+5| + \frac{11}{7}\ln|x-2| + C \,//$$

c) $\int \frac{6x+5}{x^2+x-6}\,dx$ se factoriza $\rightarrow x^2+x-6 = (x+3)(x-2)$

$$\frac{A}{(x+3)} + \frac{B}{(x-2)} = \frac{\cancel{(x+3)}(x-2)}{\cancel{(x+3)}} + \frac{(x+3)\cancel{(x-2)}}{\cancel{(x-2)}}$$

$$A(x-2) + B(x+3)$$

$x=-3$

buscar $A = \; 6(-3)+5 = A(-3-2) + B\cancel{(-3+3)}^{0}$

$-13 = A(-5) + 0$

$\frac{-13}{-5} \rightarrow \boxed{A = \frac{13}{5}}$

buscar B $\; 6(2)+5 = A\cancel{(2-2)}^{0} + B(2+3)$

$17 = 0 + B(5)$

$17 = B(5) \rightarrow \frac{17}{5} \rightarrow \boxed{\frac{17}{5} = B}$

$$\int \frac{6x+5}{(x+3)(x-2)} = \int \frac{13/5}{(x+3)} + \int \frac{17/5}{(x-2)}$$

$$\frac{13}{5}\int \frac{1}{(x+3)} + \frac{17}{5}\int \frac{1}{(x-2)}$$

$$\frac{13}{5}\ln|x+3| + \frac{17}{5}\ln|x-2| + C$$

d) $\int_0^1 \frac{2x+1}{x^2-3x+2}\,dx$ se factoriza $\rightarrow x^2-3x+2 = (x-1)(x-2)$

$$\frac{A}{(x-1)} + \frac{B}{(x-2)} = \frac{\cancel{(x-1)}(x-2)}{\cancel{(x-1)}} + \frac{(x-1)\cancel{(x-2)}}{\cancel{(x-2)}}$$

$$A(x-2) + B(x-1)$$

$x=1$

buscar $A = 2(1)+1 = A(1-2) + B\cancel{(1-1)}^{0}$

$3 = A(-1) + 0$

$\frac{3}{-1} \rightarrow \boxed{A=-3}$

$x=-2$

buscar $B = 2(2)+1 = A\cancel{(2-2)}^{0} + B(2-1)$

$5 = 0 + B(1)$

$5 = B(1) \rightarrow \frac{5}{1} \quad \boxed{B=5}$

$$\int \frac{2x+1}{(x-1)(x-2)} = \int \frac{-3}{(x-1)} + \int \frac{5}{(x-2)}$$

$$-3\int (x-1) + 5\int (x-2)$$

$$-3\ln(x-1) + 5\ln(x-2) \Big|_0^1$$

$$[-3\ln(1-1) + 5\ln(1-2)] - [-3\ln(0-1) + 5\ln(0-2)]$$

$$[-3\ln|0| + 5\ln|-1|] - [-3\ln|-1| + 5\ln|-2|]$$

indeterminado

e) $\displaystyle\int_1^3 \frac{3x-5}{x^2-4}\,dx$ se factoriza $\longrightarrow$ $x^2-4 = (x+2)(x-2)$

$$\frac{A}{(x+2)} + \frac{B}{(x-2)} \quad \frac{\cancel{(x+2)}(x-2)}{\cancel{(x+2)}} + \frac{(x+2)\cancel{(x-2)}}{\cancel{(x-2)}}$$

$$A(x-2) + B(x+2)$$

$x=-2$

buscar $A = \quad 3(-2)-5 = A(-2-2) + B\cancel{(-2+2)}^{\,0}$

$$-11 = A(-4) + 0$$

$$\frac{-11}{-4} \longrightarrow \frac{11}{4} \quad \boxed{A = \frac{11}{4}}$$

$x=2$

buscar $B = \quad 3(2)-5 = \cancel{A(2-2)}^{\,0} + B(2+2)$

$$1 = 0 + B(4)$$

$$1 = B(4) \longrightarrow \frac{1}{4} \quad \boxed{B = \frac{1}{4}}$$

$$\int \frac{3x-5}{(x-2)(x+2)} = \int \frac{11/4}{(x-2)} + \int \frac{1/4}{(x+2)}$$

$$\frac{11}{4}\int \frac{1}{x-2} + \frac{1}{4}\int \frac{1}{(x+2)}$$

$$\frac{11}{4}\ln(x-2) + \frac{1}{4}\ln(x+2) \Big/_1^3$$

$$\left[\frac{11}{4}\ln|3-2| + \frac{1}{4}\ln|3+2|\right] - \left[\frac{11}{4}\ln|1-2| + \frac{1}{4}\ln|1+2|\right]$$

$$\left[\frac{11}{4}\ln|1| + \frac{1}{4}\ln|5|\right] - \left[\frac{11}{4}\ln|-1| + \frac{1}{4}\ln|3|\right]$$

$$0 \quad + 0{,}40 \quad - \quad 0 \quad + 0{,}2$$

$$\frac{1}{5} \qquad 0{,}2\ [u^2]_{//}$$

f) $\displaystyle\int_0^2 \frac{5x-4}{x^2+x-2}\,dx$ se factoriza $\longrightarrow$ $x^2+x-2 = (x+2)(x-1)$

$$\frac{A}{(x+2)} + \frac{B}{(x-1)} = \frac{\cancel{(x+2)}(x-1)}{\cancel{(x+2)}} \quad \frac{(x+2)\cancel{(x-1)}}{\cancel{(x-1)}}$$

$$A(x-1) + B(x+2)$$

$x=-2$

buscar $A = \quad 5(-2)-4 = A(-2-1) + B\cancel{(-2+2)}^{\,0}$

$$-14 = A(-3) + 0$$

$$\frac{-14}{-3} \longrightarrow \boxed{A = \frac{14}{3}}$$

$x=1$

buscar $B = 5(1)-4 = \cancel{A(1-1)}^{\,0} + B(1+2)$

$$1 = 0 + B(3)$$

$$1 = B(3)$$

$$\frac{1}{3} \longrightarrow \boxed{B\ \frac{1}{3}}$$

$$\int \frac{5x-4}{(x-1)(x+2)} = \int \frac{14/3}{(x-1)} + \int \frac{1/3}{(x+2)}$$

$$\frac{14}{3}\int (x-1) + \frac{1}{3}\int (x+2)$$

$$\frac{14}{3}\ln|x-1| + \frac{1}{3}\ln|x+2| \Big/_0^2$$

$$\left[\frac{14}{3}\ln|2-1| + \frac{1}{3}\ln|2+2|\right] - \left[\frac{14}{3}\ln|0-1| + \frac{1}{3}\ln|0+2|\right]$$

$$0 \quad + 0.46 \qquad 0 \quad + 0.23$$

$$0.23\ [u^2]_{//}$$

Integración definida por sustitución

g) $\int_0^1 \frac{3x^2}{(x^3+1)^2}\,dx$

$u = x^3+1 \qquad du = 3x^2\,dx$

$\int \frac{du}{u^2} \to \int \frac{1}{u^2}\,du$

$\int u^{-2} \to \frac{(u)^{-1}}{-1} \to \frac{(x^3+1)^{-1}}{-1}\Big|_0^1$

$\left[\left(\frac{1^3+1}{-1}\right)^{-1} - \left(\frac{0^3+1}{-1}\right)^{-1}\right]$

$-\frac{1}{2} \quad + \quad -1 \qquad 0{,}5\,[u^2]$

h) $\int_1^4 \frac{2x}{\sqrt{x^2+3}}\,dx$

$u = x^2+3 \qquad du = 2x\,dx$

$\int \frac{du}{\sqrt{u}} \to \int \frac{1}{\sqrt{u}}\,du \to \int u^{-1/2}\,du$

$\frac{u^{1/2}}{\frac{1}{2}} \to 2u^{1/2} \to 2\sqrt{u} \to 2\sqrt{x^2+3}\Big|_1^4$

$[2\sqrt{4^2+3}] - [2\sqrt{1^2+3}]$

$2\sqrt{19} - 4 \qquad 4{,}7\,[u^2]$

$\int \cos(u)\,du = \sin(u)+C$

i) $\int_0^{\ln 3} e^{2x}\,dx$

$u = e^{2x} \qquad du = e^{2x}\cdot 2\,dx$

$\frac{1}{2}\,du = e^{2x}$

$\int \frac{1}{2} u^{2x} \to \frac{1}{2} e^{2x}\Big|_0^{\ln 3}$

Prop. log

$\left[\frac{1}{2} e^{2\ln(3)} - \frac{1}{2} e^{2(0)}\right]$

$\left[\frac{1}{2} e^{\ln(9)} - \frac{1}{2}\right]$

$\left[\frac{9}{2} - \frac{1}{2}\right] \qquad 4[u^2]$

j) $\int_0^{\frac{\pi}{4}} \cos(2x)\,dx$

$u = 2x \qquad du = 2\,dx$

$\frac{1}{2}\,du = dx$

$\int \frac{1}{2}\,du\,\cos(u)$

$\frac{1}{2}\int \cos(u)$

$\frac{1}{2}\sin(u)$

$\frac{1}{2}\sin(2x)\Big|_0^{\frac{\pi}{4}}$

angulo dentro de seno

$\left[\frac{1}{2}\sin\left(2\cdot\frac{\pi}{4}\right)\right] - \left[\frac{1}{2}\sin(0)\right]$

$\left[\frac{1}{2}\sin\left(\frac{2\pi}{4}\right)\right] - \left[\frac{1}{2}\sin(0)\right]$

$\left[\frac{1}{2}\sin\left(\frac{\pi}{2}\right)\right] - \left[\frac{1}{2}\sin(0)\right]$

$\frac{1}{2}\cdot 1 \qquad \frac{1}{2}\cdot 0$

$\frac{1}{2} - 0$

$\frac{1}{2} \qquad 0.5\,[u^2]$

k) $\int_0^1 \frac{x}{\sqrt{1-x^2}}\,dx$

$u = 1-x^2 \qquad du = -2x\,dx$

$-\frac{1}{2}\,du = x\,dx$

$\int \frac{-\frac{1}{2}du}{\sqrt{u}} \to -\frac{1}{2}\int \frac{1}{\sqrt{u}} \to -\frac{1}{2}\int u^{-1/2}$

$-\cancel{\frac{1}{2}}\,\frac{u^{1/2}}{\cancel{\frac{1}{2}}} \qquad -u^{1/2} \to -\sqrt{u} \to -\sqrt{1-x^2}\Big|_0^1$

$[-\sqrt{1-1^2}] - [-\sqrt{1-0^2}]$

$0 + 1 \qquad 1[u^2]$

l) $\int_1^e \frac{\ln x}{x}\,dx$

$u = \ln(x) \qquad du = \frac{1}{x}\,dx$

$\int u\,du \to \frac{u^2}{2} \to \frac{\ln(x)^2}{2}\Big|_1^e$

$\left[\frac{\ln(e)^2}{2} - \frac{\ln(1)^2}{2}\right]$

$\frac{1}{2} - 0$

$\frac{1}{2} = 0.5[u^2]$

Integración definida Por ParTe

m) $\int_0^1 xe^{3x}dx$

$u = x \qquad du = dx$

$v = \frac{1}{3}e^{3x} \qquad dv = e^{3x}$

$\int u\,dv = x\frac{1}{3}e^{3x} - \int \frac{1}{3}e^{3x}\,dx$

$x\frac{1}{3}e^{3x} - \frac{1}{3}\left(\frac{1}{3}e^{3x}\right)\,dx \Big/_0^1$

$x\cdot\frac{1}{3}e^{3x} - \frac{1}{9}e^{3x}\quad dx$

$\left[1\cdot\frac{1}{3}e^{3(1)} - \left(\frac{1}{9}e^{3(1)}\right)\right] - \left[0\cdot\frac{1}{3}e^{3(0)} - \left(\frac{1}{9}e^{3(0)}\right)\right]$

$4.4 \qquad\qquad 0$

$4.4\,[u^2]$ //

n) $\int_1^3 \ln x\,dx$

$u = \ln x \qquad du = \frac{1}{x}dx$

$v = x \qquad dv = dx$

$\int \ln x\,dx = \ln x\cdot x - \int x\cdot\frac{1}{x}\,dx$

$\ln x\cdot x - 1\,dx$

$\ln x\cdot x - x \Big/_1^3$

$[\ln(3)\cdot 3 - 3] - [\ln(1)\cdot 1 - 1]$

$[0.29] - [-1]$

$0.3 + 1$

$1.3\,[u^2]$ //

o) $\int_0^1 x\cos(2x)dx$ — Trigo.

$\int \cos(ax) = \frac{1}{a}\sin(ax) + c$

$\int \sin(ax) = -\frac{1}{a}\cos(ax) + c$

$u = x \qquad du = dx$

$v = \frac{1}{2}\sin(2x) \qquad dv = \cos(2x)$

$\int x\cos(2x) = x\cdot\frac{1}{2}\sin(2x) - \int\frac{1}{2}\sin(2x)\,dx$

$x\cdot\frac{1}{2}\sin(2x) - \frac{1}{2}\left(-\frac{1}{2}\cos(2x)\right)$

$x\cdot\frac{1}{2}\sin(2x) + \frac{1}{4}\cos(2x) \Big/_0^1$

$\left[1\cdot\frac{1}{2}\sin(2\cdot1) + \frac{1}{4}\cos(2\cdot1)\right] - \left[0\cdot\frac{1}{2}\sin(2\cdot0) + \frac{1}{4}\cos(2\cdot0)\right]$

$\left[\frac{1}{2}\sin(2) + \frac{1}{4}\cos(2)\right] - \left[0 + \frac{1}{4}(1)\right]$

$\left[\frac{1}{2}\sin(2) + \frac{1}{4}\cos(2)\right] - \left[\frac{1}{4}\right]$

$0.26 \quad - \quad 0.25$

$0.01\,[u^2]$ //

p) $\int_0^1 x^2\ln x\,dx$

$u = \ln x \qquad du = \frac{1}{x}dx$

$v = \frac{x^3}{3} \qquad dv = x^2\,dx$

$\int \ln x\; x^2 = \ln x\cdot\frac{x^3}{3} - \int\frac{x^3}{3}\cdot\frac{1}{x}\,dx$

$\frac{x^3}{3}\cdot x^{-1}$

$\ln x\,\frac{x^3}{3} - \int\frac{x^2}{3}$

$\ln x\,\frac{x^3}{3} - \frac{1}{3}\int x^2$

$\ln x\,\frac{x^3}{3} - \frac{1}{3}\left(\frac{x^3}{3}\right)$

$\ln x\,\frac{x^3}{3} - \frac{x^3}{9} \Big/_0^1$

$\left[\ln(1)\,\frac{1^3}{3} - \frac{1^3}{9}\right] - \left[\ln(0)\,\frac{0^3}{3} - \frac{0^3}{9}\right]$

$-\frac{1}{9}$ (indeterminado)

$$q)\ \int_0^1 x^2 e^x dx$$

integrar dos veces

$u = x^2 \qquad du = 2x\,dx$

$v = e^x \qquad dv = e^x$

$\int x^2 e^x = x^2 \cdot e^x - \int e^x\ 2x\ dx$

$\int x^2 e^x = x^2 \cdot e^x - 2\underline{\int x e^x}\ dx$

$u_2 = x \qquad du_2 = dx$

$v_2 = e^x dx \qquad dv_2 = e^x\ dx$

$\int x\ e^x = x\ e^x - \int e^x\ dx$

se juntan $\quad = x\ e^x - e^x$

$x^2 \cdot e^x - 2\ (xe^x - e^x)$

factorizado

$e^x (x^2 - 2x + 2) \Big/_0^1$

$[e^1 (1^2 - 2(1) + 2)] - [e^0 (0^2 - 2(0) + 2)]$

$2.71 \quad - \quad 2$

$0.71\ [u^2]$ //

$$r)\ \int_0^2 x e^{-x} dx$$

$u = x \qquad du = dx$

$v = -e^{-x} \quad dv = e^{-x}$

$\int x \cdot e^{-x} = x \cdot (-e^{-x}) - \int -e^{-x}\ dx$

$-xe^{-x} = -xe^{-x} + -e^{-x}$

$-xe^{-x} + -e^{-x} \Big/_0^2$

$[-2e^{-2} + -e^{-2}] - [-0e^{-0} + -e^{-0}]$

$-0.4 \qquad -1$

$-0{,}4 + 1$

$0.6\ [u^2]$ //

Teorema fundamental y Barrow

1. sea $f(x) = 3x^2 - 4x + 1$ Calcula $\int_{-1}^{2} f(x)\,dx$

$f(x) = \int_{-1}^{2} (3x^2 - 4x + 1)\ dx$

$\cancel{3}\frac{x^3}{\cancel{3}} - 4\frac{x^2}{2} + 1x \quad + C$

$x^3 - 2x^2 + 1 \Big/_{-1}^{2}$

$[2^3 - 2(2)^2 + 1(2)] - [-1^3 - 2(-1)^2 + 1(-1)]$

$[8 - 2(4) + 1(2)] - [-1^3 - 2(1) + 1(-1)]$

$[8 - 8 + 2] - [-1 - 2 - 1]$

$2 \quad - \quad -4$

$6\ [u^2]$ //

2. sea $g(x) = \int \frac{2}{x-1} - \frac{3}{x+2}$

Primitiva $\quad 2\ln|x-1| - 3\ln|x+2| \Big/_0^3$

$[2\ln|3-1| - 3\ln|3+2|] - [2\ln|0-1| - 3\ln|0+2|]$

$[2\ln|2| - 3\ln|5|] - [2\ln|1| - 3\ln|2|]$

$2\ln|2| - 3\ln|5| \qquad 0 + 3\ln|2|$

$5\ln|2| + -3\ln|5|$ //